

Agentic AI

Duration: 5 Days

A key advancement of Agentic AI is its "chaining" capability. This allows the AI to break down complex tasks into smaller, manageable steps and take a sequence of actions in response to a single request.

- Enhanced Autonomy
- Superior Problem Solving
- High Adaptability
- Deep Personalization
- Unmatched Scalability

This course provides an in-depth exploration of creating, understanding, and implementing AI agents using cutting-edge tools and methodologies. Starting with foundational knowledge of generative AI, this program guides learners through a series of practical modules to develop, refine, and innovate with AI agents. Participants will acquire hands-on experience with LangChain, LangGraph, AutoGen, and CrewAI, culminating in the construction of advanced, real-world AI applications. By the end of this course, participants will not only understand the theoretical underpinnings of AI agent development but will also be adept at practically applying these concepts in various environments.

Learning objectives

1. Develop proficiency in using LangGraph to create powerful AI agents.
2. Learn to construct and optimize multi-agent systems using AutoGen.
3. Gain skills in leveraging tools like CrewAI, AutoGen, AutoGPT to build efficient AI-driven digital workers.
4. Master the use of Agentic AI and LangChain to build database query agents and interactive applications.

Target Audience:

- Aspiring AI developers and data scientists
- Software engineers interested in AI/ML
- Technology enthusiasts with a passion for AI
- Professionals looking to upskill in advanced AI development

Prerequisites:

- Successful completion of "**Generative AI and Prompt Engineering**"
- Intermedial level knowledge of Python
- Familiarity with machine learning concepts
- A keen interest in AI and its applications

Hands on Labs:

Topics Covered

Day 1.

1. Generative AI and Prompt Engineering

- Generative AI Concepts
- Prompt and Prompt Engineering
- K-Shot Prompts
- Max Tokens
- Top-K and Top-P

Lab 1. Access the Lab Environment

Lab 2. Prompt Engineering with Command-A Model

2. The Rise of Agentic AI

- Introduction Agentic AI
- Three Waves of AI
- Agentic AI V/s Generative AI
- Core Feature of Agentic AI - Tool Calling
- Software Agents and Physical Agents
- Agentic Architecture Components

3. Coding Essentials for AI Agents

- Introduction to Python
- Programming Constructs and Syntax
- Libraries and Tools for Agentic AI
- Agentic AI Ecosystem
 - Python Module for OCI Generative AI
 - Exa, Oracle 23ai,
 - LangChain and LangGraph,
 - LlamaIndex
 - AutoGen

Lab 3. Using Exa Search Tool

Day 2

4. AI Agents – Architecture and Frameworks

- Introduction to AI agents
- Characteristics of AI Agents
- Core Components of an Agent
- How Agents Operate
- Reasoning Frameworks
- Data Stores

Lab 4. Building an AI Agent with Cohere Command A Model

5. Vectors and Embeddings

- Introduction to Oracle 23ai
- Setting up and Configuration
- Working with Vector Databases
- Understanding Embeddings
- Creating Embeddings
- Similarity Search in a Vector Database.

Lab 5. Using Oracle 23ai to Store Embeddings and Similarity Search

6. Langchain Framework

- Langchain Architecture
- Components of LangChain
- Langchain SQL Connector

Lab 6. Working with LangChain

Day 3

7. Building ReAct Agent with LangChain

- What Are ReAct Agents?
- Why Are React Agents Important?
- How Do React Agents Work?
- Creating a React Agent from Scratch
- PDF Connector as Tool.
- Website and Web Search as a Tool.

Lab 7. ReAct Agent with LangChain and Meta Llama Models

Lab 8. Building Agents with LangChain

8. Building AI Agent with LangGraph

- Overview of LangGraph
- Step-by-step project setup
- Building key Functionalities with LangGraph
- Layered Architecture for Complex Solutions
- Integrating Advanced Features
- Performance Optimization

Lab 9. Building Advanced AI Agents with LangGraph

9. RAG Based Chatbot with LangChain

- Significance Chatbots in Modern Applications
- Retrieval Augmented Generation
- Types of Chatbots

- RAG Framework
- RAG Techniques
- How RAG differs from Traditional Models
- Use Cases and Benefits of RAG in Chatbots

Lab 10. Building a Chatbot with Large Language Models

10. Building Agentic RAG with Llamaindex – Hands on

- Building Agentic RAG with Llamaindex
- Router Engine
- Tool Calling
- Building an Agent Reasoning Loop
- Building a Multi-Document Agent

Lab 11. Building Agentic RAG with Llamaindex

Day 4

11. Working with Memory

- How it Works
- Working with Memory

Lab 12. Working with Memory

12. Testing, Optimization and Evaluation

- Unit Testing and Functional Testing
- Optimizing Chatbot Performance
- Monitoring/Evaluating Chatbot Performance

Lab 13. Evaluating RAG Performance Using Langsmith

13. AI Agent with AutoGen

- Introduction to Autogen
- Setting up Autogen for projects
- Building a simple AI project
- AI Agentic Design Patterns with AutoGen
- Multi-Agent Conversation and Stand-up Comedy
- Sequential Chats and Customer Onboarding
- Reflection and Blogpost Writing
- Tool Use and Conversational Chess
- Coding and Financial Analysis
- Planning and Stock Report Generation

Lab 14. Building AI Agent with AutoGen

14. OCI Generative AI Agents Service (Theory Only)

- Overview of Generative AI Agents Service
- Your Journey in 3 Steps
- Create a Knowledge Base and Data Store
- Create the Agent
- Streamlining GenAI Agent with Streamlit

Lab 15. Building an Agent with OCI GenAI Service - (Reference Only)

15. Model Context Protocol

- What is MCP?
- Core Architecture
- MCP Client and Server
- MCP Transport Types
- Building Custom MCP Servers

Lab 16. Getting Started with MCP Server and Client

Day 5

16. Building Agents with AutoGPT

- Introduction to AutoGPT
- AutoGPT Server
- AutoGPT Frontend
- Agents and Workflows

Lab 17. Working with AutoGPT

17. Building AI Agent with CrewAI

- What is CrewAI?
- What is Composio?
- Designing an AI-Powered SQL Agent Workflow
- Prerequisites for Building the SQL Agent
- Step-by-Step Guide to Creating an SQL Agent
- Frequently Asked Questions

Lab 18(a,b). Working with CrewAI

18. Identifying and Mitigating Threats and Vulnerabilities

- Common Threats in Generative AI Systems
- Techniques to Detect and Prevent Data Leakage
- Defence Mechanisms for Model Exploits
- Credit Assignment Problem and its Impact
- Recovery Strategies from Adversarial Attacks

19. Generative AI Security Implications

- Potential Security Risks
- Importance of Data Security in the AI Context
- Principles of Data Anonymization and Encryption
- Methods for Secure Data Storage
- Techniques to Ensure Data Integrity

Lab 19. Oracle AI Optimizer